tests have passed, the API is deployed to an integration environment so that other teams can check if their integration will work successfully. I'm sure if you have experience in this, then you've run into challenges. At numerous conferences and meetups, we've asked this very question, with people stating challenges such as the following:

- Flaky dependencies, including third-party dependencies
- Environment setup and maintenance problems
- Bad seed data
- Environments down/unreliable

In an ideal world, if there were no budget constraints and a low level of complexity to spin up test environments, teams would use infrastructure as code or spin up the app within a local container. However, in our experience, development teams deal with the just-mentioned integration problems in several ways.

The first option is for teams to move away from the integration environment, add component tests that mock out the third-party dependencies, and remove the reliance on an unreliable environment. For example, teams containerize an integration environment on-demand using something like Docker. Containerized environments allow teams to test internally and in isolation, limiting the scope of their integration tests.

In the opposite direction to on-demand environments, we see teams focus on the fully integrated environment. Integration testing against a fully integrated environment gives the team the most confidence and represents a scenario closer to production. Due to the challenges around maintenance and expense with fully integrated environments, teams don't see the mocking option as a feasible solution. The confidence teams gain from a realistic environment setup outweighs the high maintenance and downtime.

However, we don't live in an ideal world because we don't all have limitless budgets or endless time to deliver the perfect solutions, so people turn to contract testing as a solution. Contract testing reduces the effort required compared with the first option and simulates the integration environment, retaining the confidence level from the second option.